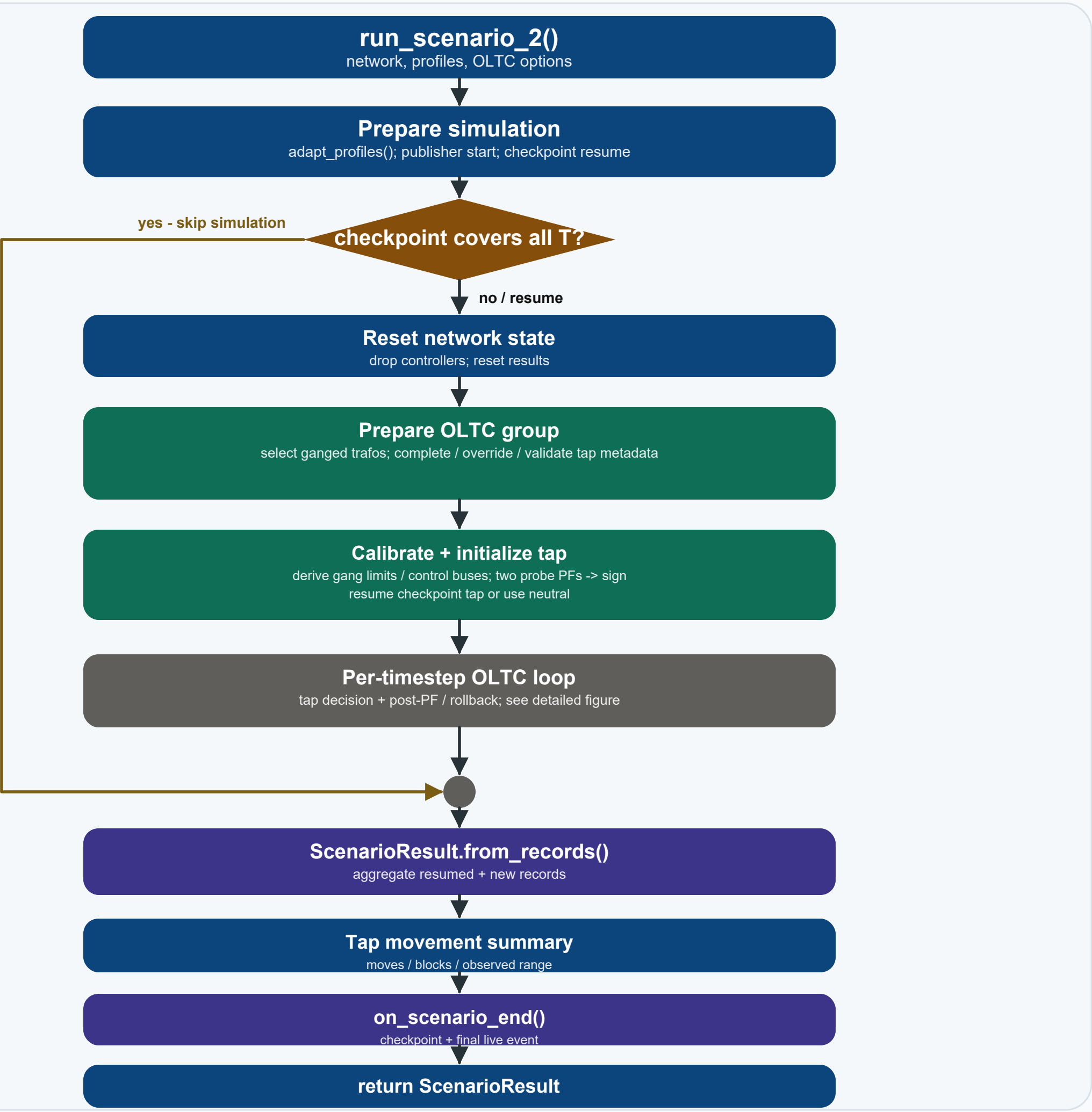




OLTC group + metadata setup

Selection Tier 1: in-service trafos with $vn_hv_kv \geq 66$ kV.

Tier 2: trafos whose hv_bus is an ext_grid slack bus.

Tier 3: all trafos at the highest available HV voltage level.

Optional completion fills only missing tap fields from benchmark defaults.

User override is then applied explicitly and may overwrite existing/default values.

Validation enforces shared neutral / side / step %, Ratio type and feasible ganged range.

Tap-sign calibration + resume state

Controlled signal = mean vm_pu across all selected LV-side busbars.

Calibration runs on `deepcopy(net)`, so working-net result tables are untouched.

Probe 1: tap_neutral; Probe 2: neutral +/-1 inside the ganged range.

Observed delta-v determines which tap direction lowers controlled voltage.

$abs(delta-v) < 1e-5$ pu is treated as an inactive / invalid tap response.

Partial resume restores current_tap from the last checkpoint record when available.

Checkpoint + lifecycle semantics

A complete checkpoint returns from resumed records before any OLTC setup or calibration.

A partial checkpoint repeats OLTC metadata validation + sign calibration, then restores tap_pos.

There is no S3-style teardown: the controlled transformer group remains at its final tap position.

Setup/calibration exceptions propagate before the timestep loop.

Pre/post/rollback runpp exceptions inside the loop are handled locally as shown in Diagram 2.

Finalization builds ScenarioResult, logs tap activity, calls `on_scenario_end()`, then returns.